



Advanced Python

Day 11

Topic: Composition (Class inside Class)

🕒 Duration: 2 Hours

🚗 Real-Life Understanding

🚗 Car example:

- Car ke andar **Engine** hota hai
- Car ke andar **Fuel Tank** hota hai
- Car ke andar **Driver** hota hai

👉 Car uses these objects → **Composition**

Definition: Ek class ke andar doosri class ka object. Relationship = **HAS-A**



Inheritance vs Composition

Inheritance

"IS-A" Relationship

Use inheritance when a class *is a specialized version* of another class.

- 👉 Dog IS-A Animal
- 👉 Manager IS-A Employee

Composition

"HAS-A" Relationship

Use composition when a class *contains or uses* another class as a part.

- 👉 Car HAS-A Engine
- 👉 Student HAS-A Address

Engine Class Example

```
class Engine: def __init__(self, horsepower): self.horsepower = horsepower def start(self):
```

Independent class that performs specific engine functions. It knows nothing about the Car.



Car Class (Composition in Action)

```
class Car: def __init__(self, brand, horsepower): self.brand = brand # COMPOSITION: Creat.
```


Car Execution

Main Script

```
# Object creation my_car = Car("Honda City", 120) # Calling methods my_car.drive() print("--")
```




Example 2: Address Class

```
class Address: def __init__(self, city, state, pincode): self.city = city self.state = state
```




Example 2: Student Class

```
class Student: def __init__(self, name, age, city, state, pincode): self.name = name self
```


Student Execution

Main Script

```
s1 = Student("Saurav", 22, "Indore", "MP", "452001") s1.show_info()
```

Output


 Naam: Sa

Summary Rule

"Class A ke andar Class B ka object =
Composition"

A HAS-A B

✓ Car HAS-A Engine

✗ Car IS-A Engine



Example 3: Phone Components

Battery Class

```
class Battery: def __init__(self, capacity): self.capacity = capacity def charge_status(self): print(f"🔋 Bat
```




Phone Class Assembly

```
class Phone: def __init__(self, brand, bat_cap, cam_mp): self.brand = brand self.battery = Battery(bat_cap) self.camera = Camera(cam_mp) def show_info:
```

📱 Phone: Samsung 🔋 Battery: 5000 mAh 📷 Photo li! Camera: 108 MP

Practice Task

Hospital System

- Create a Doctor class
- Create a Patient class
- Implement: Patient HAS-A Doctor

Expected Output:

 Patient: Ramesh  Age: 45  Doctor: Dr. Sharma  Specialization: Cardiologist



☑️☑️☑️ Revision & Checklist

🧠 Quick Revision

- ❓ Composition → IS-A ya HAS-A?
- ❓ Car(Wheel) correct hai?
- ❓ Ek class mein kitne objects ho sakte hain?

☑️ Checklist

- ✓ Composition samjha
- ✓ Examples run kiye
- ✓ Practice complete
- ✓ Difference clear



What's Next?

Day 12 → Error Handling + Custom Exceptions

Image Sources



https://img.freepik.com/free-photo/repair-tools-lying-down-engine-automobile-hood_146671-16154.jpg?semt=ais_hybrid&w=740&q=80

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